

Machine Learning

Machine learning enables computers to learn patterns from examples rather than being explicitly programmed for every situation.

Supervised Learning

Supervised learning uses labeled examples. Classification predicts categories, while regression predicts numerical values.

Unsupervised Learning

Unsupervised learning works with data that does not have target labels. Clustering is a common example.

Training and Testing

A dataset is commonly divided into training and test portions. The model learns from training data and is evaluated on unseen test data.

Overfitting

Overfitting happens when a model learns the training data too closely and performs poorly on new data.